

Boundary and Nonboundary Localization Quotients of Smooth Affine $\widehat{\mathfrak{sl}}_2$ -Modules

Unified note: the boundary, simplicity, and flow results of the living note v8
(2026-09-02)

together with the nonboundary results of *nonboundary_localization_results* (2026-09-11)

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Abstract

This note consolidates the two current strands of the project. The boundary strand (living note v8, [4]) proves $\mathcal{T}_{f_b} M_{a,b}^c(\varphi) \cong M_{a+1,b-1}^c(\psi)$, its simplicity flow, the derivation version, the iterated boundary chains, the Ω -quotient flow, and the critical-level flow with preserved θ -parameters. The nonboundary strand ([5]) studies the quotients $Q_c = \mathcal{T}_{f_c} M_{a,b}$ with $c < b$: PBW models, root-mode and Cartan profiles, weight structure, cyclicity, an explicit proper submodule at integral parameters in the deep region, a Takiff model of the basic module, annihilator preservation, and derived-functor properties. The two strands are joined by the trichotomy for arbitrary modes, the commuting-localization lemma, and the fact that iterating boundary moves fills any interval of missing modes. All statements have direct proofs; this note is not a claim of independent peer review or formal verification.

1 Setup and background

All vector spaces and tensor products are over $\mathbb{C}$. Let

$$\widehat{\mathfrak{sl}}_2 = (\mathfrak{sl}_2 \otimes \mathbb{C}[t, t^{-1}]) \oplus \mathbb{C}K, \quad \widetilde{\mathfrak{sl}}_2 = \widehat{\mathfrak{sl}}_2 \oplus \mathbb{C}d, \quad U = U(\widehat{\mathfrak{sl}}_2),$$

with $x_r = x \otimes t^r$ and the normalization

$$\begin{aligned} [e_r, f_s] &= h_{r+s} + r\delta_{r+s,0}K, & [h_r, h_s] &= 2r\delta_{r+s,0}K, \\ [h_r, e_s] &= 2e_{r+s}, & [h_r, f_s] &= -2f_{r+s}, & [d, x_r] &= rx_r. \end{aligned}$$

The e 's commute with one another, the f 's commute with one another, and K is central. Fix $a, b \in \mathbb{Z}$ with $a + b \geq 0$, set $L = a + b + 1 \geq 1$, and put

$$S_{a,b} = \text{span}\{K, h_i \ (i \geq 0), e_i \ (i > a), f_j \ (j > b)\}.$$

Fix a character $\chi : S_{a,b} \rightarrow \mathbb{C}$ with $\chi(e_i) = \chi(f_j) = 0$, $\chi(K) = \kappa$, $\chi(h_i) = \lambda_i$, and $\lambda_i = 0$ for $i > L$. These conditions make χ a Lie algebra character. Let

$$M = M_{a,b}(\chi) = U \otimes_{U(S_{a,b})} \mathbb{C}_\chi, \quad M_{a,b}^f(\chi) = \text{Ind}_{S_{a,b}}^{\widetilde{\mathfrak{sl}}_2} \mathbb{C}_\chi, \quad u = 1 \otimes 1_\chi.$$

For an injective root vector F on a module V , set

$$D_F V = U_F \otimes_U V, \quad U_F = U[F^{-1}], \quad \mathcal{T}_F V = D_F V / V.$$

More generally, for an arbitrary U -module V , define $\mathcal{T}_F V$ as the cokernel of the localization map $V \rightarrow D_F V$; then $\mathcal{T}_F V = (U_F / U) \otimes_U V$.

Fix an integer $c < b$, write

$$F = f_c, \quad d_* = b - c \geq 1, \quad Q_c = \mathcal{T}_F M = D_F M / M, \quad w_m = F^{-m}u + M \ (m \geq 1).$$

The powers of F form an Ore set because $\text{ad } F$ is locally nilpotent on U . There is no operator F^{-1} on Q_c .

A module is *smooth* if each vector is annihilated by $\mathfrak{sl}_2 \otimes t^N \mathbb{C}[t]$ for some N . An operator x is locally nilpotent if each vector is killed by a power of x , and locally finite if each vector spans a finite-dimensional $\mathbb{C}[x]$ -module. A vector v is F -torsion if $F^n v = 0$ for some n . Weights below refer to $\mathbb{C}h_0 \oplus \mathbb{C}K$.

The original induced modules satisfy the FGXZ criterion [1]:

Theorem 1.1 (FGXZ criterion). $M_{a,b}(\chi)$ is simple if and only if

$$\lambda_L \neq 0 \quad \text{and} \quad \kappa \neq -2.$$

This criterion cannot be transferred directly to Q_c .

2 PBW bases and localization normal forms

Proposition 2.1 (Left-normal PBW model and pole filtration). Choose a PBW order with $F = f_c$ first, followed by the free families

$$f_j \ (j \leq b, j \neq c), \quad e_i \ (i \leq a), \quad h_r \ (r < 0),$$

and finally a basis of $S_{a,b}$. Let $\mathcal{X}$ denote the ordered monomials in the three middle families, including the empty monomial. Then

$$\begin{aligned} M &: \{F^q Xu : q \geq 0, X \in \mathcal{X}\}, \\ D_F M &: \{F^q Xu : q \in \mathbb{Z}, X \in \mathcal{X}\}, \\ Q_c &: \{F^{-m} Xu + M : m \geq 1, X \in \mathcal{X}\} \end{aligned}$$

are bases. In particular, Q_c is nonzero and countably infinite-dimensional. The operator F on Q_c is surjective and locally nilpotent, and

$$\ker F^r = \text{span}\{F^{-m} Xu + M : 1 \leq m \leq r, X \in \mathcal{X}\} \quad (r \geq 1).$$

Every nonzero $\mathfrak{g}$ -submodule of Q_c meets $\ker F$ nontrivially.

Proof. The first basis is the induced-module PBW theorem, and it makes M free over $\mathbb{C}[F]$. Localization gives the second basis: spanning follows by moving denominators to the left, and linear independence by multiplying a putative relation by a common power of F . Quotienting gives the third. The kernel formulas are then immediate; for the last assertion, apply to a nonzero vector of a submodule the highest power of F that does not kill it. $\square$

Lemma 2.2 (Commutation and generation by pure fractions). For $s \in U$ and $m \geq 1$,

$$sF^{-m} = \sum_{j \geq 0} \binom{m+j-1}{j} F^{-m-j} (\text{ad } F)^j(s), \quad (1)$$

a finite sum. Moreover, $Q_c = \sum_{m \geq 1} U w_m$.

Proof. For $m = 1$, commute s past F^{-1} repeatedly; the process terminates because $\text{ad } F$ is locally nilpotent, and induction on m gives the binomial coefficient. For the generation assertion, give a numerator PBW word the degree $\rho = 2\#e + \#h$, with f -factors of degree zero. Reordering does not increase this degree, and every nonzero application of $\text{ad } F$ lowers it strictly. Thus

$$X w_m = F^{-m} Xu + M + \text{a finite sum of terms of strictly smaller numerator degree.}$$

Induct on $\rho(X)$, simultaneously for all denominators m . $\square$

Lemma 2.3 (Right-normal localized PBW). Let $F = f_r$ with $r \leq b$. Choose a PBW order for $M_{a,b}^c(\chi)$ in which f_r is the *last* free variable, and let $\mathcal{X}_r$ denote the remaining ordered monomials. Then

$$\{XF^q u : X \in \mathcal{X}_r, q \in \mathbb{Z}_{\geq 0}\}, \quad \{XF^q u : X \in \mathcal{X}_r, q \in \mathbb{Z}\}, \quad \{XF^{-m} u + M : X \in \mathcal{X}_r, m \geq 1\}$$

are bases of $M_{a,b}^c(\chi)$, $D_F M_{a,b}^c(\chi)$, and $\mathcal{T}_F M_{a,b}^c(\chi)$, respectively. Correspondingly, $\{d^k XF^q u : k \geq 0, X \in \mathcal{X}_r, q \in \mathbb{Z}\}$ is a basis of $D_F M_{a,b}^f(\chi)$, and in the quotient only $q < 0$ remains.

Proof. ad F is locally nilpotent on U and on $U(\widetilde{\mathfrak{sl}}_2)$, and $[F, d] = -rF$, so $\{F^m : m \geq 0\}$ is an Ore set. The first basis is PBW with f_r last. In the localization, the finite commutation formulas move negative powers to the right of every ordered monomial. Linear independence follows on the associated graded, where the localization is the localization of a polynomial ring at $\text{gr } F$. Placing d first gives the last assertions. $\square$

3 The boundary case

Set

$$F = f_b, \quad E = e_{a+1}, \quad H = h_{a+b+1} = h_L.$$

Define $\psi : S_{a+1,b-1} \rightarrow \mathbb{C}$ by

$$\psi(K) = \chi(K), \quad \psi(h_0) = \chi(h_0) + 2, \quad \psi(h_i) = \chi(h_i) \ (i > 0),$$

and zero on the root vectors of $S_{a+1,b-1}$. Thus ψ differs from χ only in the h_0 -eigenvalue.

Lemma 3.1 (ψ is a character). The displayed assignment defines a Lie algebra character $\psi : S_{a+1,b-1} \rightarrow \mathbb{C}$.

Proof. The only bracket that can fail is $[e_i, f_j] = h_{i+j} + i\delta_{i+j,0}K$ with $i > a+1, j > b-1$. Then $i+j \geq a+b+2 = L+1$, so the central term vanishes and $\psi(h_{i+j}) = \lambda_{i+j} = 0$. The brackets $[h_i, e_j], [h_i, f_j]$ land in annihilated tails, and $[h_i, h_j] = 2i\delta_{i+j,0}K$ with $i, j \geq 0$ has value zero. $\square$

Lemma 3.2 (Cyclic vector). The vector $\bar{u} = F^{-1}u + M \in \mathcal{T}_F M$ satisfies

$$x\bar{u} = \psi(x)\bar{u} \quad (x \in S_{a+1,b-1}).$$

Proof. For $j > b-1$: if $j = b$, then $f_b \bar{u} = u + M = 0$; if $j > b$, then $f_j u = 0$. From $h_i F^{-1} = F^{-1}h_i + 2F^{-2}f_{i+b}$ one gets $h_0 \bar{u} = (\lambda_0 + 2)\bar{u}$, since $2F^{-2}f_b u = 2F^{-2}F u = 2\bar{u}$, and $h_i \bar{u} = \lambda_i \bar{u}$ for $i > 0$. For $j > a+1$,

$$e_j F^{-1} = F^{-1}e_j - F^{-2}h_{j+b} - 2F^{-3}f_{j+2b},$$

with $j+b > L$, so $h_{j+b} u = 0$, and $j+2b > b$, so $f_{j+2b} u = 0$. Hence $e_j \bar{u} = 0$, and $K\bar{u} = \kappa \bar{u}$. $\square$

Theorem 3.3 (Boundary localization). If

$$\lambda_L = \chi(h_L) \neq 0,$$

then there is a natural $\widehat{\mathfrak{sl}}_2$ -module isomorphism

$$\boxed{\mathcal{T}_{f_b} M_{a,b}^c(\chi) \cong M_{a+1,b-1}^c(\psi),}$$

sending the standard generator u' of the target to

$$\bar{u} = f_b^{-1}u + M.$$

The statement is independent of the level κ .

Proof. By Lemma 3.2 and the universal property of induction, there is a unique $\widehat{\mathfrak{sl}}_2$ -module map

$$\Phi : M_{a+1,b-1}^c(\psi) \longrightarrow \mathcal{T}_F M, \quad u' \longmapsto \bar{u}.$$

Let $\mathcal{X}$ be the ordered monomials in e_i ($i \leq a$), h_j ($j < 0$), f_k ($k \leq b-1$), and choose the source PBW order with E last. Then $\{X E^p u' : X \in \mathcal{X}, p \geq 0\}$ is a basis of $M_{a+1,b-1}^c(\psi)$, and by Lemma 2.3 $\{X F^{-p-1} u + M : X \in \mathcal{X}, p \geq 0\}$ is a basis of the quotient. The relations

$$[E, F] = H, \quad (\text{ad } F)^2(E) = -2f_{a+2b+1},$$

together with $Eu = 0$, $Hu = \lambda_L u$, and $f_{a+2b+1}u = 0$, give

$$E F^s u = s \lambda_L F^{s-1} u \quad (s \in \mathbb{Z}),$$

and in particular

$$E^p F^{-1} u = (-1)^p p! \lambda_L^p F^{-p-1} u.$$

Therefore

$$\Phi(X E^p u') = (-1)^p p! \lambda_L^p X F^{-p-1} u + M,$$

so Φ sends a PBW basis bijectively onto nonzero scalar multiples of a PBW basis and is an isomorphism. $\square$

Remark 3.4 (Surjection plus simplicity shortcut). If only simplicity is needed, one may avoid the base comparison. The chain $E F^{-m} u = -m \lambda_L F^{-m-1} u$ gives all pure fractions in the image of Φ , and Lemma 2.2 gives surjectivity. If $\kappa \neq -2$, then by Theorem 1.1 the source $M_{a+1,b-1}^c(\psi)$ is simple, because $\psi(h_L) = \lambda_L \neq 0$ and $\psi(K) = \kappa \neq -2$. A nonzero surjection from a simple module is an isomorphism.

Corollary 3.5 (Simplicity of the boundary quotient). If $\lambda_L \neq 0$ and $\kappa \neq -2$, then $\mathcal{T}_{f_b} M_{a,b}^c(\chi)$ is simple: it is isomorphic to $M_{a+1,b-1}^c(\psi)$, whose two irreducibility parameters are

$$\psi(h_{a+b+1}) = \lambda_L, \quad \psi(K) = \kappa.$$

Remark 3.6 (Why not the full localization). The full localization $D_F M$ contains M as a nonzero submodule, so it cannot be simple. The object carrying the standard-module structure is the quotient $\mathcal{T}_F M = D_F M / M$. Twisted localizations keep F invertible and are likewise not standard induced modules.

Proposition 3.7 (Boundary b -function). With $F = f_b$, $E = e_{a+1}$,

$$E F^s u = s \chi(h_L) F^{s-1} u \quad (s \in \mathbb{Z}),$$

equivalently $E F^{s+1} u = (s+1) \chi(h_L) F^s u$, so the monic b -polynomial is

$$b(s) = s + 1, \quad \text{or, with parameters,} \quad b_\chi(s) = (s+1) \chi(h_L).$$

The condition $\lambda_L \neq 0$ is exactly the absence of obstruction in the negative-power direction.

Proof. This is the finite expansion $E F^s = F^s E + s F^{s-1} H - 2 \binom{s}{2} F^{s-2} f_{a+2b+1}$ evaluated on u . See [2, Section 5] for the analogous resonance for free-field modules. $\square$

4 Arbitrary modes: trichotomy and the gap invariant

Proposition 4.1 (Arbitrary-mode trichotomy). Let $M = M_{a,b}^c(\chi)$ and consider $F = f_r$.

- (i) If $r > b$, then F is locally nilpotent on M , hence $D_F M = 0$: there is no nontrivial localization.
- (ii) If $r = b$ and $\chi(h_L) \neq 0$, then $\mathcal{T}_{f_b} M \cong M_{a+1,b-1}^c(\psi)$ by Theorem 3.3.
- (iii) If $r < b$, then $D_F M \neq 0$, but $\mathcal{T}_F M$ is not isomorphic to any standard module $M_{a',b'}^c(\chi')$: the set of locally nilpotent negative root modes is $\{r\} \cup \{j : j > b\}$, which has a gap, while standard modules have full upper tails $\{j : j > b'\}$.

The same trichotomy holds for $M_{a,b}^f(\chi)$. Case (iii) is developed in depth in Sections 5–7.

Proof. For $r > b$, $Fu = 0$ and $F^m(xu) = (\text{ad } F)^m(x)u$ vanishes for $m \gg 0$, so F is locally nilpotent on M . For $r < b$, Lemma 2.3 gives the localized basis; F is locally nilpotent on the quotient, f_j for $j \leq b$, $j \neq r$ acts as an injective free polynomial variable, and f_j for $j > b$ inherits local nilpotence from M . $\square$

Proposition 4.2 (The negative-root profile). On Q_c ,

$$\begin{array}{l|l} j = c & f_j \text{ is surjective and locally nilpotent,} \\ j \leq b, j \neq c & f_j \text{ is injective and not surjective,} \\ j > b & f_j \text{ is locally nilpotent.} \end{array}$$

Consequently the set of locally nilpotent negative-root modes is exactly $\{c\} \cup \{j : j > b\}$.

Proof. The first assertion is Proposition 2.1. For $j \leq b$, $j \neq c$, f_j is multiplication by a free polynomial variable in the commuting algebra model. For $j > b$, use $f_j u = 0$, local nilpotence of $\text{ad } f_j$, and $[f_j, F] = 0$. $\square$

Corollary 4.3 (Nonisomorphism and Hom vanishing). For $c < b$, Q_c is not isomorphic to any standard module $M_{a',b'}^c(\psi)$. For fixed M and distinct $c, c' \leq b$,

$$\text{Hom}_{\mathfrak{g}}(T_{f_c} M, T_{f_{c'}} M) = 0.$$

Proof. The local-nilpotence set of a standard module is a half-line. For the Hom assertion, f_c is locally nilpotent on the source and injective on the target. $\square$

Proposition 4.4 (Smoothness and weight structure). The module Q_c is smooth, K acts by κ , and h_0 acts semisimply with

$$\text{Supp}_{h_0} Q_c = \lambda_0 + 2\mathbb{Z}.$$

Every nonzero weight space is countably infinite-dimensional. The module has no nonzero finite-dimensional submodules or quotients, and is not integrable for the full affine algebra.

Proof. If X has r_e e -factors and r_f f -factors, then

$$h_0(F^{-m}Xu + M) = (\lambda_0 + 2m + 2r_e - 2r_f)(F^{-m}Xu + M).$$

All weights in the stated coset occur; appending powers of h_{-1} gives infinitely many independent vectors of each weight. Smoothness follows from the finite commutation formulas

$$\begin{aligned} h_r F^{-m} &= F^{-m} h_r + 2m F^{-m-1} f_{r+c}, \\ e_r F^{-m} &= F^{-m} e_r - m F^{-m-1} (h_{r+c} + r \delta_{r+c,0} K) - m(m+1) F^{-m-2} f_{r+2c}, \end{aligned}$$

by taking $r, r+c, r+2c$ sufficiently large. Surjectivity and local nilpotence of F exclude nonzero finite-dimensional quotients, and a finite-dimensional smooth $\mathfrak{g}$ -module is trivial; trivial submodules are excluded because f_{c+1} is injective. A free root mode f_j , $j \leq b$, $j \neq c$, excludes integrability. $\square$

5 Positive Cartan modes and localization depth

Proposition 5.1 (Strict growth of pole order). For $1 \leq i \leq d_*$, $\zeta \in \mathbb{C}$, and $0 \neq v \in Q_c$,

$$\text{pole}_F((h_i - \zeta)v) = \text{pole}_F(v) + 1,$$

where pole order means the largest denominator exponent in the PBW expansion. In particular,

$$\ker p(h_i) = 0 \quad (0 \neq p \in \mathbb{C}[t]).$$

Thus h_i has no nonzero locally finite vectors, eigenvectors, or generalized eigenvectors.

Proof. Write the top pole term of v as $F^{-m}v_m + M$, $v_m \neq 0$. The term $F^{-m}h_iv_m$ has pole order at most m , and the coefficient of pole order $m+1$ is $2mf_{c+i}v_m \pmod{FM}$. Since $c+i \leq b$ and $c+i \neq c$, multiplication by f_{c+i} is injective on M/FM , so this coefficient is nonzero and cannot be cancelled by lower terms or by $-\zeta v$. Iteration shows that $v, h_iv, \dots, h_i^n v$ have distinct pole orders. $\square$

Corollary 5.2. For every standard induced module,

$$\text{Hom}_{\mathfrak{g}}(M_{a',b'}(\psi), Q_c) = 0.$$

More generally, Q_c has no nonzero character vector for any Lie subalgebra containing h_1 .

Proposition 5.3 (High Cartan modes). If $i > d_*$, then $h_i w_m = \lambda_i w_m$ for every $m \geq 1$, and $h_i - \lambda_i$ is locally nilpotent on all of Q_c .

Proof. Since $c+i > b$, the correction term in the h -commutation kills u . For a general smooth module, if $i > 0$ and $h_i w = \lambda w$, then $(h_i - \lambda)^N Xw = (\text{ad } h_i)^N(X)w$; for large N every surviving term contains a root factor of arbitrarily high index, which annihilates w . Apply this to all w_m and use Lemma 2.2. $\square$

Consequently the positive Cartan modes with no nonzero locally finite vectors are precisely $h_1, \dots, h_{b-c}$: a second intrinsic way to read off the localization depth.

6 The basic module $Q_0 = T_{f_0}M_{-1,1}$

Here $a = -1$, $b = 1$, $c = 0$, and $L = d_* = 1$. All statements allow arbitrary $\lambda_0, \lambda_1, \kappa$. The pure fractions satisfy

$$\begin{aligned} f_0 w_1 &= 0, & f_0 w_m &= w_{m-1} \quad (m > 1), \\ h_0 w_m &= (\lambda_0 + 2m)w_m, \end{aligned} \tag{2}$$

$$e_0 w_m = -m(\lambda_0 + m + 1)w_{m+1}, \tag{3}$$

$$h_1 w_m = \lambda_1 w_m + 2mf_1 w_{m+1},$$

$$e_1 w_m = -m\lambda_1 w_{m+1} - m(m+1)f_1 w_{m+2}.$$

Unlike boundary localization at f_1 , the pure-fraction recursion is controlled by λ_0 , not by λ_1 .

Proposition 6.1 (Exact cyclicity statement). The module Q_0 is cyclic for all parameters, and

$$Q_0 = Uw_1 \iff \lambda_0 \notin \{-2, -3, -4, \dots\}.$$

If $\lambda_0 = -s$ with $s \geq 2$, then $Q_0 = Uw_s$, while Uw_1 is a proper submodule.

Proof. If λ_0 is not a negative integer at most -2 , every coefficient in (3), starting at $m = 1$, is nonzero, and induction gives $w_m \in Uw_1$ for all m ; Lemma 2.2 then gives $Q_0 = Uw_1$. If $\lambda_0 = -s$, the coefficient vanishes precisely at $m = s - 1$; starting at w_s , applications of e_0 produce all w_m with $m \geq s$, and applications of f_0 produce $w_{s-1}, \dots, w_1$, so $Q_0 = Uw_s$. But $e_0^{s-1}w_1 = 0$, so $Uw_1 \subseteq G$, while $w_s \notin G$, with G as in Proposition 6.2. $\square$

Proposition 6.2 (The e_0 -locally nilpotent part). Let

$$G = \{v \in Q_0 : e_0^N v = 0 \text{ for some } N \geq 1\}.$$

This is a $\mathfrak{g}$ -submodule, and

$$\begin{cases} 0 \neq G \neq Q_0, & \lambda_0 \in \mathbb{Z}, \\ G = 0, & \lambda_0 \notin \mathbb{Z}. \end{cases}$$

Thus Q_0 is reducible for every integral λ_0 . For nonintegral λ_0 , e_0 is injective; simplicity is not asserted.

Proof. Set $E = e_0$. Since $\text{ad } E$ is locally nilpotent on U , the expansion

$$E^n(sv) = \sum_{r=0}^n \binom{n}{r} (\text{ad } E)^r(s) E^{n-r}v$$

proves $UG \subseteq G$.

Suppose first $p = \lambda_0 \in \mathbb{Z}$. Choose $k \geq \max(1, p + 2)$, and put $x = f_0$, $y = f_1$, $\eta = \lambda_1$. The subspace $\mathbb{C}[x, y]u$ is stable under e_0, h_0, f_0 , with

$$e_0(x^j y^\ell u) = j(p - j + 1 - 2\ell)x^{j-1}y^\ell u + \ell\eta x^j y^{\ell-1}u. \quad (4)$$

Define

$$v_k = \sum_{j=0}^k a_j x^j y^{k-j} u, \quad a_0 = 1, \quad a_j = -\frac{(k-j+1)\eta}{j(p-2k+j+1)} a_{j-1}. \quad (5)$$

All denominators are nonzero, since $p - 2k + j + 1 \leq p - k + 1 < 0$. The coefficient of $x^{j-1}y^{k-j}u$ in $e_0 v_k$ is

$$j(p - 2k + j + 1)a_j + (k - j + 1)\eta a_{j-1} = 0,$$

so $e_0 v_k = 0$, and $h_0 v_k = \nu v_k$ with $\nu = p - 2k \leq -2$. The rank-one commutation formula gives

$$e_0(f_0^{-m} v_k) = -m(\nu + m + 1)f_0^{-m-1} v_k,$$

whose coefficient vanishes at $m = -\nu - 1$. In Q_0 ,

$$f_0^{-1} v_k + M = f_1^k w_1 \neq 0, \quad e_0^{-\nu-1}(f_1^k w_1) = 0,$$

so $G \neq 0$. On the other hand, for $m \geq \max(1, -p)$ every coefficient in the successive applications of (3) to w_m is nonzero, so $w_m \notin G$: the submodule G is proper.

Now suppose $\lambda_0 \notin \mathbb{Z}$ and $G \neq 0$. Since G is h_0 -stable and h_0 is semisimple, it contains a nonzero weight vector; applying a power of e_0 produces a nonzero $v' \in \ker e_0$. Write $h_0 v' = \nu v'$ and choose the least $N \geq 1$ with $f_0^N v' = 0$. The $\mathfrak{sl}_2$ -identity gives

$$0 = e_0 f_0^N v' = N(\nu - N + 1)f_0^{N-1} v',$$

so by minimality $\nu = N - 1 \in \mathbb{Z}$, contradicting $\nu \in \lambda_0 + 2\mathbb{Z}$. $\square$

Remark 6.3 (What this submodule does and does not prove). The submodule G is the largest submodule on which e_0 is locally nilpotent, and is locally finite for $\langle e_0, h_0, f_0 \rangle \cong \mathfrak{sl}_2$. The quotient Q_0/G is nonzero and e_0 acts injectively on it. Neither simplicity of G or Q_0/G , nor length two or indecomposability of Q_0 , has been proved.

Proposition 6.4 (An exact Takiff model). The subspace

$$V = \text{span}\{f_1^k w_m : m \geq 1, k \geq 0\} \cong \mathbb{C}[x, x^{-1}, y]/\mathbb{C}[x, y]$$

is stable under the nonnegative current algebra $\mathfrak{sl}_2[t]$, with all modes of degree at least two acting by zero; it is therefore a module for the Takiff algebra $\mathfrak{sl}_2[t]/t^2\mathfrak{sl}_2[t]$. Under $f_1^k w_m \leftrightarrow x^{-m}y^k$,

$$\begin{aligned} f_0 &= x, & f_1 &= y, \\ h_0 &= \lambda_0 - 2x\partial_x - 2y\partial_y, \\ h_1 &= \lambda_1 - 2y\partial_x, \\ e_0 &= \lambda_0\partial_x + \lambda_1\partial_y - x\partial_x^2 - 2y\partial_x\partial_y, \\ e_1 &= \lambda_1\partial_x - y\partial_x^2. \end{aligned}$$

These are exact formulas, not a degree truncation. The subspace V is not a submodule for the full affine algebra (negative Cartan modes leave it).

7 General nonboundary indices: deep and shallow regions

7.1 Spectral-flow normalization

For $k \in \mathbb{Z}$, define the automorphism

$$\sigma_k(e_i) = e_{i+k}, \quad \sigma_k(f_i) = f_{i-k}, \quad \sigma_k(h_i) = h_i + k\delta_{i,0}K, \quad \sigma_k(K) = K.$$

Our twist convention is $x \cdot_{\text{new}} v = \sigma_k(x)v$. Tracking the inducing conditions and the localization element gives

$$(Q_c)^{\sigma_{-c}} \cong T_{f_0} M_{a+c, b-c}(\chi'), \quad \chi'(h_0) = \mu := \lambda_0 - c\kappa, \quad (6)$$

with $\chi'(K) = \kappa$ and $\chi'(h_i) = \lambda_i$ for $i > 0$. This is an isomorphism *after twisting*, not an untwisted isomorphism. The useful normalized coordinates are

$$A = a + c, \quad B = b - c = d_*, \quad A + B = L - 1, \quad \mu = \lambda_0 - c\kappa.$$

Proposition 7.1 (The deep region). Suppose $c \leq -a - 1$, equivalently $d_* \geq L$. Put

$$E = e_{-c}, \quad H = h_0 - cK, \quad F = f_c.$$

These elements form an $\mathfrak{sl}_2$ -triple, $Eu = 0$, $Hu = \mu u$, and

$$Ew_m = -m(\mu + m + 1)w_{m+1}.$$

The following statements hold without assuming $\lambda_L \neq 0$:

- (i) $Q_c = Uw_1$ if and only if $\mu \notin \{-2, -3, \dots\}$. If $\mu = -s$, $s \geq 2$, then $Q_c = Uw_s \neq Uw_1$.
- (ii) The subspace

$$G_c = \{v \in Q_c : E^N v = 0 \text{ for some } N \geq 1\}$$

is a $\mathfrak{g}$ -submodule. It is nonzero and proper when $\mu \in \mathbb{Z}$, and zero when $\mu \notin \mathbb{Z}$.

- (iii) The operator E acts injectively on the nonzero quotient Q_c/G_c .

Proof. The commutator $[e_{-c}, f_c] = h_0 - cK$ gives the rank-one triple; the deep-region assumption is exactly $-c > a$, so $Eu = 0$. Set $Y_b = f_b$. On $\mathbb{C}[F, Y_b]u$,

$$E(F^j Y_b^\ell u) = j(\mu - j + 1 - 2\ell)F^{j-1}Y_b^\ell u + \ell\lambda_{d_*}F^j Y_b^{\ell-1}u, \quad (7)$$

which is (4) with (p, η, x, y) replaced by $(\mu, \lambda_{d_*}, F, Y_b)$. When $\mu \in \mathbb{Z}$, the highest vector (5) with these replacements gives $0 \neq f_b^k w_1 \in G_c$ with $E^{2k-\mu-1}(f_b^k w_1) = 0$; if $d_* > L$, then $\lambda_{d_*} = 0$ and the vector simply equals $f_b^k u$. Pure fractions with $m \geq \max(1, -\mu)$ are not E -locally nilpotent, so G_c is proper. For $\mu \notin \mathbb{Z}$, the least- N argument of Proposition 6.2 forces an integral highest weight, a contradiction, so $G_c = 0$. Injectivity on Q_c/G_c follows from $Ev \in G_c \Rightarrow v \in G_c$. $\square$

Proposition 7.2 (The shallow region). Suppose $-a \leq c < b$, equivalently $1 \leq d_* < L$. Then $E = e_{-c}$ acts injectively on Q_c , for every value of μ . If $\lambda_L \neq 0$, the vector w_1 generates Q_c .

Proof. In this region e_{-c} belongs to the free PBW complement, so $Eu = 0$ cannot be used. Instead set $Y = e_{L-c}$. Since $L - c > a$ and $L + c > b$,

$$Yu = 0, \quad [Y, F] = h_L, \quad f_{L+c}u = 0,$$

and there is no central term in the middle identity because $L \geq 1$. The e -commutation formula therefore yields

$$Yw_m = -m\lambda_L w_{m+1},$$

so $\lambda_L \neq 0$ gives cyclicity via Lemma 2.2.

For injectivity of E , use numerator degree $\rho = 2\#e + \#h$: on the highest-degree part of a nonzero PBW expansion in Q_c , left multiplication by E adds a free e_{-c} -factor, increasing the degree by two, while all commutator terms have degree at most one above the original. In the associated graded PBW space the highest term is multiplication by a free polynomial variable, hence injective, and it cannot cancel. $\square$

8 Extensions, annihilators, and derived functors

Proposition 8.1 (The natural extension is nonsplit). The short exact sequence

$$0 \longrightarrow M \longrightarrow D_F M \longrightarrow Q_c \longrightarrow 0$$

does not split. In fact,

$$\mathrm{Hom}_{\mathfrak{g}}(Q_c, D_F M) = \mathrm{Hom}_{\mathfrak{g}}(Q_c, M) = \mathrm{Hom}_{\mathfrak{g}}(M, Q_c) = 0.$$

If M is simple, it is the unique simple submodule of $D_F M$ and an essential submodule; consequently $D_F M$ is indecomposable.

Proof. The source Q_c is F -torsion, whereas F is injective on $D_F M$ and on M ; this proves the first two vanishings and excludes a section. The third is Corollary 5.2. For $0 \neq v \in D_F M$, a sufficiently large power of F sends v to a nonzero element of M . $\square$

Proposition 8.2 (Annihilator preservation). In the ordinary universal enveloping algebra,

$$\mathrm{Ann}_U M = \mathrm{Ann}_U D_F M = \mathrm{Ann}_U Q_c.$$

Proof. Let $\Delta = \text{ad } F$. The two-sided ideal $\text{Ann}_U M$ is Δ -stable, so (1) shows that every element of $\text{Ann}_U M$ annihilates $D_F M$; the inclusion $M \subseteq D_F M$ gives equality of the first two annihilators, and this common ideal is contained in $\text{Ann}_U Q_c$. For the reverse inclusion, let $s \in \text{Ann}_U Q_c$, and choose $t \geq 0$ with $\Delta^{t+1}(s) = 0$. For $v \in M$ and $m \geq 1$, $sF^{-m}v \in M$. Multiplying by F^{m+t} gives

$$P_v(m) := \sum_{j=0}^t \binom{m+j-1}{j} F^{t-j} \Delta^j(s)v \in F^{m+t}M,$$

a vector-valued polynomial in m . Its finitely many coefficients have bounded nonnegative F -exponents, so for all sufficiently large m the left side lies in a space intersecting $F^{m+t}M$ trivially; hence $P_v \equiv 0$. Evaluating at $m = 0$ gives $F^t s v = 0$, because $\binom{j-1}{j} = 0$ for $j \geq 1$; injectivity of F gives $s v = 0$. $\square$

Proposition 8.3 (Functorial and derived properties). For an arbitrary U -module V ,

$$\begin{aligned} T_F V &= (U_F/U) \otimes_U V, \\ L_1 T_F(V) &\cong \{v \in V : F^n v = 0 \text{ for some } n \geq 0\}, \\ L_j T_F(V) &= 0 \quad (j \geq 2). \end{aligned}$$

The functor is right exact and is exact on short exact sequences whose three terms are F -torsion-free. In particular,

$$D_F Q_c = 0, \quad T_F Q_c = 0, \quad L_1 T_F(Q_c) \cong Q_c.$$

Proof. Ore localization is flat as a right U -module, so $0 \rightarrow U \rightarrow U_F \rightarrow U_F/U \rightarrow 0$ is a flat resolution of U_F/U . Tensoring with V identifies its degree-one homology with $\ker(V \rightarrow D_F V)$, the F -torsion submodule, and there is no higher homology. $\square$

9 Further localization, the boundary chains, and iteration

Proposition 9.1 (Commuting quotient localizations). Let $F = f_c$ and $G = f_j$ with distinct $c, j \leq b$, and write $D_{F,G}M$ for simultaneous localization. There are natural isomorphisms

$$T_G(T_F M) \cong \frac{D_{F,G}M}{D_F M + D_G M} \cong T_F(T_G M).$$

The analogous statement holds for any finite family of distinct free negative-root modes, independently of their order.

Proof. The modes commute, their localizations commute, and PBW makes M free in the two independent polynomial variables F, G ; inside $D_{F,G}M$,

$$D_F M \cap D_G M = M.$$

Moreover, G is injective on $T_F M$. Localizing the exact sequence defining $T_F M$ at G gives $D_G(T_F M) \cong D_{F,G}M/D_G M$, and quotienting by the image of $T_F M$ gives the middle expression. Its PBW basis consists of terms with both exponents strictly negative. $\square$

Corollary 9.2 (Filling the interval of missing modes). Assume $\lambda_L \neq 0$. Applying one quotient localization at each of

$$f_c, f_{c+1}, \dots, f_b$$

in any order yields

$$M_{a+b-c+1, c-1}(\chi^{(b-c+1)}), \quad \chi^{(b-c+1)}(h_0) = \lambda_0 + 2(b-c+1),$$

with all other character values unchanged.

Proof. By Proposition 9.1, arrange the operations in the order $f_b, f_{b-1}, \dots, f_c$. At each step the next mode is a boundary mode for the resulting standard module, and Theorem 3.3 applies. The value of $L = a + b + 1$ and the nonzero parameter λ_L are unchanged throughout. $\square$

For the basic example,

$$T_{f_1}(T_{f_0}M_{-1,1}(\chi)) \cong T_{f_0}(T_{f_1}M_{-1,1}(\chi)) \cong M_{1,-1}(\chi^{(2)}), \quad \lambda_1 \neq 0.$$

These are iterated *quotient* localizations; one must divide by the sum of the relevant partial localizations, and simplicity of the final module does not imply simplicity of the first quotient.

Theorem 9.3 (Iterated boundary chain). Assume $\chi(h_L) \neq 0$. For $k \geq 1$ define $\chi^{(k)}$ by

$$\chi^{(k)}(K) = \chi(K), \quad \chi^{(k)}(h_0) = \chi(h_0) + 2k, \quad \chi^{(k)}(h_i) = \chi(h_i) \quad (i > 0).$$

Then

$$\boxed{\mathcal{T}_{f_{b-k+1}} \cdots \mathcal{T}_{f_{b-1}} \mathcal{T}_{f_b} M_{a,b}^c(\chi) \cong M_{a+k,b-k}^c(\chi^{(k)}),} \quad (8)$$

with the rightmost functor acting first, using the same nondegeneracy condition $\chi(h_L) \neq 0$ at every step. In particular, for $M_{-1,N}^c(\chi)$ and $n \leq N$, taking $k = N - n + 1$,

$$\boxed{\mathcal{T}_{f_n} \mathcal{T}_{f_{n+1}} \cdots \mathcal{T}_{f_N} M_{-1,N}^c(\chi) \cong M_{N-n,n-1}^c(\chi^{(N-n+1)}).} \quad (9)$$

10 Adjoining the derivation

On $M_{a,b}^f(\chi) \cong \mathbb{C}[d] \otimes M_{a,b}^c(\chi)$, the $\widetilde{\mathfrak{sl}}_2$ -action is the semidirect one: for x_r and $p(d) \in \mathbb{C}[d]$,

$$d(p(d) \otimes v) = dp(d) \otimes v, \quad x_r(p(d) \otimes v) = p(d - r) \otimes x_r v.$$

In particular, $F^{-1}(p(d) \otimes v) = p(d + b) \otimes F^{-1}v$ for $F = f_b$.

Proposition 10.1 (Localization commutes with adjoining d). Let X be an $F = f_b$ -torsion-free $\widetilde{\mathfrak{sl}}_2$ -module. Then there are natural $\widetilde{\mathfrak{sl}}_2$ -module isomorphisms

$$D_F(\text{Ind}_{\widetilde{\mathfrak{sl}}_2}^{\widetilde{\mathfrak{sl}}_2} X) \cong \text{Ind}_{\widetilde{\mathfrak{sl}}_2}^{\widetilde{\mathfrak{sl}}_2}(D_F X), \quad \boxed{\mathcal{T}_F(\text{Ind}_{\widetilde{\mathfrak{sl}}_2}^{\widetilde{\mathfrak{sl}}_2} X) \cong \text{Ind}_{\widetilde{\mathfrak{sl}}_2}^{\widetilde{\mathfrak{sl}}_2}(\mathcal{T}_F X).}$$

As vector spaces, $\mathcal{T}_F(\mathbb{C}[d] \otimes X) \cong \mathbb{C}[d] \otimes \mathcal{T}_F X$.

Proof. Set $A = U(\widehat{\mathfrak{sl}}_2)$, $B = U(\widetilde{\mathfrak{sl}}_2)$. Since $[d, F] = bF$, the derivation $\text{ad } d$ extends to $D_F A$, and PBW gives $D_F B \cong B \otimes_A D_F A$. Hence $D_F(B \otimes_A X) \cong B \otimes_A D_F X$. Since B is free as a right A -module, $B \otimes_A -$ is exact, and applying it to $0 \rightarrow X \rightarrow D_F X \rightarrow \mathcal{T}_F X \rightarrow 0$ gives the quotient isomorphism. $\square$

Theorem 10.2 (Boundary localization with the derivation). If $\chi(h_L) \neq 0$, then

$$\boxed{\mathcal{T}_{f_b} M_{a,b}^f(\chi) \cong M_{a+1,b-1}^f(\psi).}$$

In the PBW normal form $M^f \cong \mathbb{C}[d] \otimes M^c$, the isomorphism is termwise

$$d^k X E^p u' \mapsto (-1)^p p! \lambda_L^p d^k X F^{-p-1} u + M_{a,b}^f(\chi).$$

The iterated chains (8)–(9) hold verbatim for M^f .

Proposition 10.3 (Nonboundary version with the derivation). There is a natural $\tilde{\mathfrak{sl}}_2$ -module isomorphism

$$\mathcal{I}(Q_c) \cong T_{f_c} \mathcal{I}(M), \quad \mathcal{I}(V) = \operatorname{Ind}_{\tilde{\mathfrak{sl}}_2}^{\tilde{\mathfrak{sl}}_2} V \cong \mathbb{C}[d] \otimes V,$$

given explicitly by

$$\alpha(F^{-m}(P(d) \otimes v)) = P(d + mc) \otimes F^{-m}v,$$

with inverse $P(d) \otimes F^{-m}v \mapsto F^{-m}(P(d - mc) \otimes v)$. If $0 \neq N \neq Q_c$ is a $\mathfrak{g}$ -submodule, then $0 \neq \mathcal{I}(N) \neq \mathcal{I}(Q_c)$.

Proof. On $\mathcal{I}(M)$, the target of $\mathcal{I}(M) \rightarrow \mathcal{I}(D_F M)$ has an F -invertible action, with

$$F(P(d) \otimes v) = P(d - c) \otimes Fv, \quad F^{-1}(P(d) \otimes v) = P(d + c) \otimes F^{-1}v.$$

The universal property of localization extends the map to $\alpha : D_F \mathcal{I}(M) \rightarrow \mathcal{I}(D_F M)$ with the displayed formula, which respects equivalent fractions. The algebra $U(\tilde{\mathfrak{sl}}_2)$ is free as a right U -module, so $\mathcal{I}$ is exact; quotienting identifies the cokernels. Exactness and faithfulness of $\mathcal{I}$ prove the last assertion. $\square$

11 Non-critical simple flow with the Casimir

Assume $\kappa \neq -2$. The Casimir element Ω of $\tilde{\mathfrak{sl}}_2$ commutes with $U(\tilde{\mathfrak{sl}}_2)$; to simplify notation take the representative $P(\Omega) = 1$ and write

$$L_{a,b}(\chi; \mu) := M_{a,b}^f(\chi) / (\Omega - \mu) M_{a,b}^f(\chi).$$

Theorem 11.1 (Non-critical simple flow). Assume

$$\chi(h_L) \neq 0, \quad \kappa \neq -2.$$

Then f_b is injective on $L_{a,b}(\chi; \mu)$, and

$$\boxed{\mathcal{T}_{f_b} L_{a,b}(\chi; \mu) \cong L_{a+1,b-1}(\psi; \mu).}$$

Both sides are simple $\tilde{\mathfrak{sl}}_2$ -modules, and the Casimir parameter μ does not change. More generally,

$$\mathcal{T}_{f_{b-k+1}} \cdots \mathcal{T}_{f_b} L_{a,b}(\chi; \mu) \cong L_{a+k,b-k}(\chi^{(k)}; \mu).$$

Proof. By [1, formula (4.1)], at level $\kappa \neq -2$ the restriction of $L_{a,b}(\chi; \mu)$ to $\widehat{\mathfrak{sl}}_2$ is isomorphic to $M_{a,b}^c(\chi)$; hence f_b remains injective. Since Ω commutes with $F = f_b$, exactness of Ore localization gives

$$\mathcal{T}_F(M^f / (\Omega - \mu)M^f) \cong (\mathcal{T}_F M^f) / (\Omega - \mu)(\mathcal{T}_F M^f).$$

Theorem 10.2 turns the right side into $M_{a+1,b-1}^f(\psi) / (\Omega - \mu)M_{a+1,b-1}^f(\psi) = L_{a+1,b-1}(\psi; \mu)$. The irreducibility parameters of [1, Theorem 4.1] are unchanged, so both sides are simple. Iterate. $\square$

Remark 11.2 (Why one cannot stop at M^f). At non-critical level, $(\Omega - \mu)M^f$ is a proper submodule of M^f , so the raw module M^f and the raw quotient $\mathcal{T}_{f_b} M^f \cong M_{a+1,b-1}^f(\psi)$ are generally not simple. The complete simple flow is

$$\boxed{\text{boundary localization quotient} + \Omega\text{-central reduction},}$$

and the two operations commute, with μ preserved.

12 Critical level: the θ -parameters are preserved

Throughout this section $\kappa = \chi(K) = -2$ and $q := L = a + b + 1$. For the reduced case $(a, b) = (-1, q)$, the critical central modes used in [1] are the Segal–Sugawara modes $T(q - i)$.

Proposition 12.1 (Spectral flow on the Sugawara modes). Let σ_k be as in Section 7. In the completed enveloping algebra,

$$\sigma_k(T(m)) = T(m) + \frac{k}{2}(K+2)h_m + \frac{k^2}{4}K(K+2)\delta_{m,0}.$$

Consequently, at the critical level $K = -2$,

$$\sigma_k(T(m)) = T(m)$$

as operators on any smooth module.

Proof. Write $T(m)$ as the combination $\frac{1}{4}(2, 2, 1)$ of the three normal-ordered sums of [1, formula (2.3)]. For $k > 0$, spectral flow moves the normal-ordering cut of the first sum from $i > 0$ to $i > -k$, producing $kh_m + \frac{k(k-1)}{2}K\delta_{m,0}$; the second sum produces $kh_m + \frac{k(k+1)}{2}K\delta_{m,0}$; the third produces $2kKh_m + k^2K^2\delta_{m,0}$. Inserting the coefficients $\frac{1}{4}(2, 2, 1)$ gives the formula; negative k follows from $\sigma_{k+\ell} = \sigma_k\sigma_\ell$, and $K = -2$ gives the invariance. $\square$

For the critical character, define

$$Z_i^{a,b} := \sigma_{-a-1}(T(q-i)), \quad i \in \mathbb{N},$$

and let $I_\theta^{a,b}$ be the ideal of the critical center generated by $Z_i^{a,b} - \theta_i$. Put

$$\overline{M}_{a,b}^c(\chi; \theta) := M_{a,b}^c(\chi)/I_\theta^{a,b}M_{a,b}^c(\chi), \quad \overline{M}_{a,b}^f(\chi; \theta) := \text{Ind}_{\widehat{\mathfrak{sl}}_2}^{\widehat{\mathfrak{sl}}_2} \overline{M}_{a,b}^c(\chi; \theta).$$

By Proposition 12.1, at level -2

$$Z_i^{a,b} = T(q-i) = Z_i^{a+1,b-1},$$

so the boundary move does not change θ .

Theorem 12.2 (Critical simple flow). Assume

$$\chi(K) = -2, \quad \chi(h_q) \neq 0.$$

Then f_b is injective on $\overline{M}_{a,b}^c(\chi; \theta)$, and

$$\mathcal{T}_{f_b} \overline{M}_{a,b}^c(\chi; \theta) \cong \overline{M}_{a+1,b-1}^c(\psi; \theta),$$

both sides being simple $\widehat{\mathfrak{sl}}_2$ -modules. The same holds with f in place of c throughout, both sides being simple $\widehat{\mathfrak{sl}}_2$ -modules.

Proof. The critical irreducibility theorem of [1] makes $\overline{M}_{a,b}^c(\chi; \theta)$ simple. The F -torsion vectors form a submodule because $\text{ad } F$ is locally nilpotent; the critical PBW basis contains the nonzero vectors $f_q^m \bar{u}$, and twisting back by σ_{a+1} shows that f_b is not locally nilpotent on the whole module. Hence f_b is injective. Since each $T(m)$ commutes with $\widehat{\mathfrak{sl}}_2$ and with F , exactness of Ore localization gives

$$\mathcal{T}_F \left(M_{a,b}^c / I_\theta^{a,b} M_{a,b}^c \right) \cong \frac{\mathcal{T}_F M_{a,b}^c}{I_\theta^{a,b} (\mathcal{T}_F M_{a,b}^c)}.$$

Theorem 3.3 and $Z_i^{a,b} = Z_i^{a+1,b-1} = T(q-i)$ identify the right side with $\overline{M}_{a+1,b-1}^c(\psi; \theta)$. The f -version follows from Proposition 10.1, with simplicity from [1, Theorem 4.2]. $\square$

Iterating,

$$\mathcal{T}_{f_{b-k+1}} \cdots \mathcal{T}_{f_b} \overline{M}_{a,b}^\bullet(\chi; \theta) \cong \overline{M}_{a+k, b-k}^\bullet(\chi^{(k)}; \theta), \quad \bullet \in \{c, f\},$$

with θ unchanged along the entire chain.

13 Vertex-algebra viewpoint

By smoothness and the fixed action of K , each Q_c is a weak module for the universal affine vertex algebra $V^\kappa(\mathfrak{sl}_2)$; see [1, Section 2]. Descending to its simple quotient requires a separate check that the defining vertex ideal annihilates the module; smoothness alone does not imply this.

14 Open problems

1. **Nonintegral simplicity in the basic case.** — **Resolved** by the addendum (2026-09-15): under $\lambda_1 \neq 0, \kappa \neq -2$,

$$T_{f_0} M_{-1,1}(\chi) \text{ simple} \iff \lambda_0 \notin \mathbb{Z},$$

via the rank-one containment argument (Theorem on the containment of all proper submodules).

2. **The corresponding deep-region criterion.** — **Resolved** by the same addendum: under $\lambda_L \neq 0, \kappa \neq -2, d_* \geq L$,

$$\mathcal{T}_{f_c} M_{a,b} \text{ simple} \iff \mu = \lambda_0 - c\kappa \notin \mathbb{Z}.$$

3. **The integral-parameter submodule structure.** — **Partially resolved; internal structure of G_c still open.** The addendum proves that G_c is the unique maximal submodule, Q_c/G_c is simple, and the extension is nonsplit. The further questions (simplicity of G_c , whether it is the socle, its Jordan–Hölder length) were addressed in a note of 2026-09-22; an independent audit of that date (audits/2026-09-22/) found counterexamples to several of its claims (E-action formula, primitive-space description, the weight-filtration), which have been withdrawn in the corrected notes `gc_structure_advisor_en` and `gc_structure_research_cn`. The internal classification of G_c remains open; what is retained is the transverse $\mathfrak{sl}_2$ -decomposition and the corrected primitive recursion.
4. **Simplicity in the shallow region.** Here e_{-c} is injective for all μ , so the deep-region integral obstruction disappears. Does simplicity of the original module suffice?
5. **Parameter isomorphisms and degenerations.** The level κ , support $\lambda_0 + 2\mathbb{Z}$, negative-root local-nilpotence set, and positive-Cartan local-finiteness profile give necessary invariants; for $i > d_*$, the generalized eigenvalue λ_i is an additional invariant. A complete classification, especially for $\lambda_L = 0$ or $\kappa = -2$, remains unresolved.
6. **Critical-level central reduction.** At $\kappa = -2$, study the completed center, its central reductions, and their compatibility with localization; Proposition 8.2 does not settle completed-center questions.
7. **Multiple missing modes and twisted localization.** Proposition 9.1 suggests a family indexed by finite sets of free modes. One may also study the generalized conjugation

$$\Theta_\xi(s) = \sum_{j \geq 0} \binom{\xi}{j} (\text{ad } F)^j(s) F^{-j}, \quad \xi \in \mathbb{C},$$

on $D_F M$, where the sum is finite for each $s \in U$. This requires F to be invertible; it is not directly an operation on Q_c .

15 Sources and verification record

The original induced modules, their simplicity criterion, spectral-flow automorphisms, and the smooth-module vertex-algebra correspondence are drawn from [1]; [2] provides the localization comparison for free-field modules. The boundary isomorphism and the derivation-induction framework were established in the earlier project notes [3, 4], and the nonboundary results in [5].

Machine checks. The Takiff action formulas of Section 6 were checked with 5,866 exact rational evaluations (`work/check_nonboundary_takiff.py`), including the six generators' bracket relations, the highest-vector recurrence at integral parameters, explicit locally nilpotent vectors, and pure-fraction resonance, with $\lambda_1 = 0$ included and no polynomial-degree truncation. Independently, in the verification round of 2026-09-02 the key formulas underlying Sections 2, 3, 10, and 12 were checked symbolically with three engines (Python/SymPy, Mathematica 14.3, GAP 4.15.1), 76 checks in total, all passing: the localization formulas as a consistent Laurent action for all $s, t \in \mathbb{Z}$ (both sides are polynomials of degree at most three in s, t), the closed commutation forms, the σ_k automorphism on all defining relations, the character property of ψ , the cyclic-vector relations, the chain $E^p F^{-1} u = (-1)^p p! \lambda_L^p F^{-p-1} u$ for a grid of (a, b) , the derivation shifts, and the composition law plus $K = -2$ invariance of the spectral-flow formula. Two boundary facts were also accepted into the Danus LLM-review fact graph (fact IDs d6bf6fef62dc8971, b388fabb701946d3). No Lean/Coq formalization is claimed, and the paper-level theorems (the FGXZ criterion, the critical irreducibility statement, and the completed-center computations) remain external inputs.

References

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- [2] V. Futorny, D. Grantcharov, and R. A. Martins, *Localization of free field realizations of affine Lie algebras*, arXiv:1404.7148v2 (2015). [arXiv full text](#).
- [3] Earlier project working notes, *Boundary localization: advisor note* (September 2026) and *Boundary localization and derivation extension* (September 8, 2026).
- [4] Project living note, *Smooth representations of affine Kac–Moody algebras: PBW bases, irreducibility, and localization tasks*, version v8 (2026-09-02).
- [5] Project note, *Nonboundary localization quotients of smooth affine $\mathfrak{sl}_2$ -modules* (2026-09-11).